

Prompt Ladder — from a lazy prompt to a reusable one

Alex (Xiaoyu) Ji · FlyRank · track: Machine Learning · task: prioritizing which SEO pages to refresh (FlyRank content_refresh dataset) · 2026-07-12

One layer per version. Every version tied to **exactly one** named layer. Outputs compared side by side, not from memory.

Baseline (embarrassing on purpose)

Prompt:

Here's an SEO dataset. Analyze it and tell me what's interesting.

Output (excerpt):

Great question! This is a rich dataset with lots of potential. A few interesting things to explore:

- **Traffic:** look at which pages get the most pageviews and sessions.
- **CTR:** click-through rate shows how engaging your titles are — pages above 76% CTR are doing great.
- **Length:** longer articles tend to rank better, so consider adding more words.
- **Engagement:** higher engagement rate is better.

Overall there's lots of opportunity here — let me know if you'd like me to dive into any area!

Vague, hedgy, a menu instead of an answer, misreads CTR ($0.76 = 0.76\%$, not 76%), and hands me SEO clichés ("longer ranks better"). Could describe any dataset on earth.

V1 — layer added: a clearer goal

Prompt: *Here's an SEO dataset. Rank which pages I should refresh first to recover declining traffic.*

Output (excerpt):

To prioritize refreshes: (1) find pages trending down, (2) among those keep the ones with high impressions (worth saving), (3) drop brand-new pages that haven't stabilized. Sort by decline × traffic — declining pages with high impressions are your best candidates. Update their content and metadata.

Four notes

- *What changed in the prompt:* replaced "tell me what's interesting" with one goal — rank pages to refresh first.
 - *What improved in the output:* it stopped offering a menu of directions and committed to a single ranked task (decline + demand); the answer finally has a spine.
 - *What still failed:* still generic — it never named a real column or told me how to actually compute the ranking. Same paragraph would fit any SEO tool.
 - *What I'd try next:* feed it the real schema so it stops guessing.
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V2 — layer added: real context (actual columns + the dataset's gotchas)

Prompt adds: *Columns include impressions_90d, clicks_90d, ctr, avg_position, days_since_last_update, content_age_days, trend_direction, impressions_prev_30d... Gotchas: ctr/engagement_rate/ai_traffic_pct are x100 percentages (0.76 = 0.76%); avg_position = 0 means "no data," not rank zero; is_declining_label is derived from trend_direction, so trend_direction and trend_pct must never be model inputs.*

Output (excerpt):

A refresh candidate = trend_direction == 'down' **and** still has demand (high impressions_90d) **and** is stale (days_since_last_update large). I'll exclude trend_direction/trend_pct from any score — they define the label. I'll treat avg_position == 0 as missing, not as rank #1. And since ctr is a x100 percent, "low CTR" means roughly below ~1.0, not below 0.01.

Four notes

- *What changed in the prompt:* pasted the real column list and three dataset gotchas.
- *What improved in the output:* it stopped inventing column names and **stopped misreading CTR** — it correctly read ctr as a x100 percent and **excluded the leakage columns** from the score. The advice became specific to *this* dataset. Biggest single jump of the ladder.
- *What still failed:* it's a prose recipe, not something a person could act on — no ranking produced, no format.
- *What I'd try next:* say who reads this, so the language fits them.

V3 — layer added: a defined audience (a non-technical content editor)

Prompt adds: *Write this for a content editor who doesn't read code or stats.*

Output (excerpt):

Here's the plan in plain terms: we'll flag pages that are slipping in Google but still get seen by lots of people — those are worth your time first. We'll skip brand-new pages and pages nobody visits. For each flagged page you'll get one suggested action: refresh the content, rewrite the title/description, or just keep an eye on it. No code or stats needed.

Four notes

- *What changed in the prompt:* defined the audience as a non-technical content editor.
- *What improved in the output:* it dropped the jargon (leakage, x100) and reframed everything as **editor actions** (refresh / rewrite title / monitor) instead of column math.
- *What still failed (over-corrected):* it threw the **reasons** out with the jargon — the editor now gets actions with no "why," so nothing to trust. Softer, but thinner.
- *What I'd try next:* pin a format that keeps the reason next to each action.

V4 — layer added: a specified output format (ranked table + reason codes + confidence) — ⚠️ this one made it worse

Prompt adds: *Return a ranked table: Rank · Page · Action · Reason codes · Confidence.*

Output (excerpt):

Rank	Page	Action	Reasons	Confidence
1	content_304f48230142	Refresh + rewrite title	declining, high impressions, low CTR	High
2	content_a1fb4e703a9e	Refresh	declining, stale, strong demand	High
3	content_9b21c7f0e4aa	Monitor	declining, low demand	Medium

Four notes

- *What changed in the prompt:* specified the output format — a ranked table with reason codes and a confidence column.
- *What improved in the output:* structure. It's now a scannable queue an editor can work top-down, with a reason and confidence beside each row.
- **What still failed — and got WORSE:** asking for a *filled-in* table made the model **fabricate** specific `content_ids`, ranks, and confidence labels it has no way to know from a prompt. The format made it look authoritative while inventing data — **more dangerous than the vague baseline**, because it's confidently wrong. (This is the honest "a layer backfired" moment.)
- *What I'd try next:* add verification rules — only use values it can actually derive; mark any demo row as illustrative.

V5 — layer added: verification requirements

Prompt adds: *Only use values you can actually compute or that I give you. Do NOT invent page IDs, numbers, or confidence levels. If you show the format with placeholder data, label it ILLUSTRATIVE. Use careful language ("appears to be declining," "candidate for review" — never "will recover" or "proven"). State your assumptions.*

Output (excerpt):

Assumptions: "visible" = `impressions_90d ≥ 100`; "stale" = `days_since_last_update ≥ 180`. I don't have the computed scores, so the table below is **ILLUSTRATIVE (format only — placeholder rows)**:
 I Rank | Page (id) | Action | Reasons | Confidence |
 I --|--|--|--|--|
 I 1 | «from your data» | Refresh — candidate for review | declining_with_demand, low_ctr_visible | «from model» |

These are **observed / directional** candidates, not guarantees a refresh will recover a page — a declining page may be consolidating or seasonal.

Four notes

- *What changed in the prompt:* added verification requirements — no fabricated data, label illustrative rows, careful language, state assumptions.
- *What improved in the output:* it **stopped inventing data** — labeled the sample table ILLUSTRATIVE, refused fake confidence numbers, and switched to careful wording ("candidate for review," not "will recover"). Trustworthy for the first time.
- *What still failed:* nothing broke — but the output is now a template + method, not a finished list, because it correctly *can't* rank real pages without the computed scores. Honest, but the real ranking still has to happen in code.
- *What I'd try next:* pair this prompt with the pipeline's actual scores (paste them in) so it fills real rows under these same guardrails.

What the ladder taught me (one line)

More prompt is not always better: **V2 (real context) earned the most, and V4 (format) actively backfired** by turning structure into fabrication — until V5 (verification) put a guardrail on it. The layer that mattered wasn't the fanciest; it was the one that killed the current output's worst failure.

Final reusable prompt (a stranger on my track can use this)

```
You are helping prioritize which SEO pages to review first for a content refresh,
using the FlyRank content-performance dataset.

GOAL: rank pages that are declining in traffic but still worth saving, so a content
editor knows what to work on first.

AUDIENCE: a non-technical content editor – plain language, concrete actions, no stats jargon.

CONTEXT (respect these dataset rules):
- ctr, engagement_rate, ai_traffic_pct are x100 percentages (0.76 means 0.76%).
- avg_position = 0 means "no data," not rank #1.
- is_declining_label is derived from trend_direction; NEVER use trend_direction or
  trend_pct as inputs to a score (that's leakage).
- "visible" = impressions_90d ≥ 100; "stale" = days_since_last_update ≥ 180.

OUTPUT FORMAT: a ranked table – Rank | Page (id) | Action (Refresh / Rewrite title / Expand / Monitor) | Reason codes |
Confidence.

VERIFICATION (do not skip):
- Use ONLY values I provide or that are computable from the data. Do NOT invent page
  ids, numbers, or confidence levels.
- If you show the format without real data, label the table ILLUSTRATIVE.
- Use careful language: "appears to be declining," "candidate for review" – never
  "will recover" or "proven." State your assumptions up front.

INPUT: [paste the page rows or the pipeline's computed scores here]
```